Chapter 6. Explainable Models: Subset Selection, Shrinkage and Dimension Reduction

Lecture 6-3 (best subset, stepwise, C_p/AIC/BIC/CV, ridge, lasso, high-dimensional theory, PCR, PLS)

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1 Three Classes of Methods

- **Subset selection**: identify a subset of the p predictors related to the response and fit least squares on it.
- **Shrinkage (regularisation)**: fit all p predictors but shrink the coefficients toward zero relative to least squares; this reduces variance and can also select variables.
- **Dimension reduction**: project the p predictors onto an M-dimensional subspace ($M < p$) and regress on the M projections by least squares.

2 Subset Selection

2.1 Best subset selection

Algorithm: best subset selection

1. Let $\mathcal{M}_0$ be the null model (no predictors; predicts the sample mean).
2. For $k = 1, \ldots, p$: fit all $\binom{p}{k}$ models with exactly k predictors and call the best (smallest RSS, or largest R^2) $\mathcal{M}_k$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using a chosen score (CV error, C_p, AIC, BIC, adjusted R^2).

Within step 2, models of the same size can be compared by RSS; across sizes (step 3), RSS always favours the largest model, so a score that estimates test error is needed. Best subset examines 2^p models: accurate but computationally infeasible for large p, and statistically prone to overfitting (the huge search space lets a model fit noise).

2.2 Stepwise selection

Algorithm: forward stepwise selection

1. Let $\mathcal{M}_0$ be the null model.
2. For $k = 0, \ldots, p-1$: consider the $p-k$ models that add one predictor to $\mathcal{M}_k$; the best of them is $\mathcal{M}_{k+1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ by CV error, C_p, AIC, BIC or adjusted R^2.

Algorithm: backward elimination

1. Let $\mathcal{M}_p$ be the full model.
2. For $k = p, p-1, \ldots, 1$: consider the k models that drop one predictor from $\mathcal{M}_k$; the best is $\mathcal{M}_{k-1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$.

Both search $1 + \sum_{k=0}^{p-1}(p-k) = 1 + p(p+1)/2$ models (e.g. 211 instead of $2^{20} \approx 10^6$ for $p = 20$). Neither is guaranteed to find the best subset: the best two-variable model need not contain the

best one-variable predictor. Backward elimination needs $n > p$ so the full model can be fitted ($\hat{\beta} = (X^\top X)^{-1} X^\top Y$ requires $X^\top X$ invertible); forward selection works even with $n < p$ (up to $\mathcal{M}_{n-1}$) and is often the only viable subset method when p is very large.

2.3 Choosing the optimal model

Each procedure returns a sequence $\mathcal{M}_0, \dots, \mathcal{M}_p$; we choose the size $\hat{k}$. Indirect estimates of test error adjust the training error:

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \text{AIC, BIC (Ch. 2)}, \qquad R^2_{\text{adj}},$$

with d the number of parameters and $\hat{\sigma}^2$ estimated from the full model.

Theorem: C_p is unbiased for the in-sample prediction error

For a least-squares fit $\hat{y} = Hy$ with $\mathrm{tr}H = d$, $y = \mu + \varepsilon$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and new responses y^{new} at the same inputs,

$$\mathbb{E}\Big[\frac{1}{n}\|y^{\text{new}} - \hat{y}\|^2\Big] = \mathbb{E}\Big[\frac{1}{n}\|y - \hat{y}\|^2\Big] + \frac{2d\sigma^2}{n}.$$

Proof. Expanding both sides, the difference is $\frac{2}{n}\sum_i \mathrm{Cov}(y_i, \hat{y}_i) = \frac{2}{n}\mathrm{trCov}(y, Hy) = \frac{2}{n}\sigma^2 \mathrm{tr}H = \frac{2d\sigma^2}{n}$. ■

The direct alternative is the validation-set or CV error for each $\mathcal{M}_k$: it estimates test error directly, makes fewer assumptions about the true model, and does not need $\hat{\sigma}^2$. On the Credit data, $\sqrt{\text{BIC}}$, validation error and CV error all flatten after about four to six predictors. A **one-standard-error rule** picks the smallest model whose estimated test error is within one SE of the minimum.

3 Shrinkage Methods

Instead of least squares on a subset, fit all p predictors with a technique that constrains or regularises the coefficients, shrinking them toward zero. Shrinking can significantly reduce their variance.

3.1 Ridge regression

$$\hat{\beta}^R = \arg\min_{\beta} \Big\{ \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij}\Big)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \Big\} = \arg\min_{\beta} \{\mathrm{RSS} + \lambda \|\beta\|_2^2\},$$

with tuning parameter $\lambda \geq 0$ (the intercept is not penalised). Ridge seeks coefficients that fit well (small RSS) but the *shrinkage penalty* is small only when $\beta_1, \dots, \beta_p$ are near zero. $\lambda = 0$ gives least squares; $\lambda \to \infty$ gives $\hat{\beta}^R \to 0$. λ is chosen by CV. For centred data $\hat{\beta}^R = (X^\top X + \lambda I)^{-1} X^\top Y$.

Scaling. Least squares is scale equivariant: multiplying X_j by c divides $\hat{\beta}_j$ by c, so $\hat{\beta}_j X_j$ is unchanged. Ridge estimates can change substantially, because the penalty treats all coefficients alike. Standardise first:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}}.$$

On the Credit data, the standardised ridge coefficients shrink smoothly toward zero as λ grows (or as $\|\hat{\beta}^R_\lambda\|_2 / \|\hat{\beta}\|_2$ decreases from 1 to 0).

3.2 Why ridge improves over least squares

In a simulation with $n = 50$ and $p = 45$ (all coefficients nonzero), the squared bias increases and the variance decreases with λ, and the test MSE is minimised at an intermediate λ, well below that of least squares.

Theorem: bias and variance of ridge

For $Y = X\beta + \varepsilon$, $\varepsilon \sim (0, \sigma^2 I)$, and $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top Y$,

$$\mathbb{E}\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top X\beta, \qquad \text{Bias} = -\lambda (X^\top X + \lambda I)^{-1}\beta,$$

$$\mathrm{Var}(\hat\beta_\lambda) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}, \qquad \text{MSE} = \|\text{Bias}\|^2 + \text{trVar}.$$

Moreover there is always some $\lambda > 0$ with $\mathrm{MSE}(\hat\beta_\lambda) < \mathrm{MSE}(\hat\beta_{\mathrm{OLS}})$ (Hoerl and Kennard, 1970).

Proof. The mean and variance follow from linearity in Y; $(X^\top X + \lambda I)^{-1} X^\top X - I = -\lambda (X^\top X + \lambda I)^{-1}$. With $X^\top X = V D^2 V^\top$ and $\theta = V^\top \beta$,

$$\mathrm{MSE}(\lambda) = \sum_{j=1}^{p} \frac{\sigma^2 d_j^2 + \lambda^2 \theta_j^2}{(d_j^2 + \lambda)^2}.$$

Differentiating, $\mathrm{MSE}'(\lambda) = \sum_j \frac{2 d_j^2 (\lambda \theta_j^2 - \sigma^2)}{(d_j^2 + \lambda)^3}$, which equals $-2\sigma^2 \sum_j d_j^{-4} < 0$ at $\lambda = 0$. So MSE decreases when moving from OLS ($\lambda = 0$) to small positive λ. ∎

As $\lambda \to \infty$ the bias grows and the variance vanishes; a proper λ balances the two, achieving lower MSE than OLS. The gain is largest when some d_j are small (collinearity) or p is close to n.

3.3 The lasso

Ridge keeps all p predictors in the model, which hampers interpretation when p is large. The lasso fixes this:

$$\hat\beta^L = \arg\min_\beta \{\mathrm{RSS} + \lambda \|\beta\|_1\}, \qquad \|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

Like ridge it shrinks toward zero, but the ℓ_1 penalty forces some coefficients to be **exactly zero** when λ is large enough. Like best subset, the lasso performs variable selection and gives **sparse** models; λ is chosen by CV. On the Credit data, coefficients enter the lasso path one at a time as λ decreases.

3.4 Constrained formulations and geometry

Ridge and lasso solve

$$\min_\beta \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_j \beta_j x_{ij}\Big)^2 \quad \text{subject to} \quad \sum_j \beta_j^2 \le s \quad \text{resp.} \quad \sum_j |\beta_j| \le s,$$

for some $s = s(\lambda)$ (Lagrange duality). Best subset is the same with $\sum_j \mathbb{I}(\beta_j \neq 0) \le s$. The RSS contours are ellipses around $\hat\beta_{\mathrm{OLS}}$; the lasso constraint is a diamond with corners on the axes, so the first contact often occurs at a corner where some $\beta_j = 0$; the ridge constraint is a disc with no corners, so contact is typically off the axes.

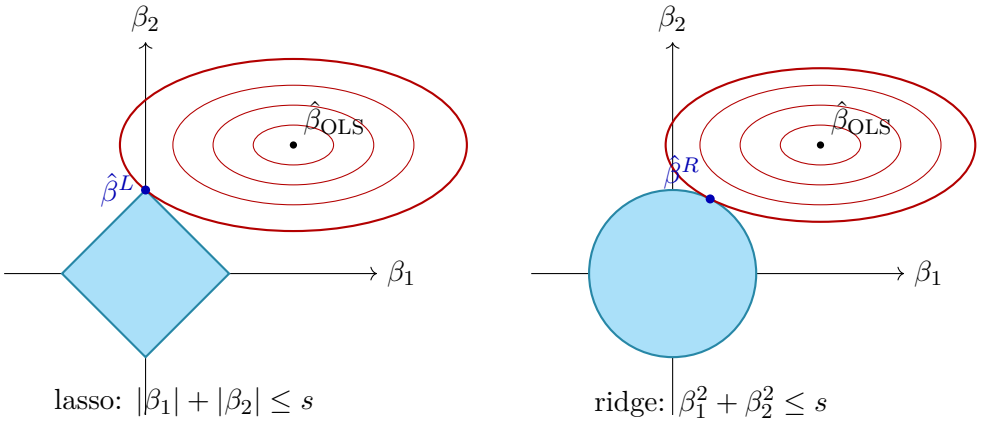


Figure 1: Contours of RSS (red ellipses) around $\hat{\beta}_{\text{OLS}}$ and the constraint regions (blue). The solution is the first point where an ellipse touches the region. The diamond's corners lie on the axes, so the lasso often lands on a corner ($\hat{\beta}_1 = 0$ here); the disc has no corners, so ridge shrinks both coefficients without zeroing either. (Contact points computed for the drawn ellipses.)

Theorem: orthonormal design

If $n = p$ and $X = I$ (no intercept), then for RSS $+ \lambda\cdot$penalty:

$$\hat{\beta}_j^R = \frac{y_j}{1+\lambda}, \qquad \hat{\beta}_j^L = \begin{cases} y_j - \lambda/2, & y_j > \lambda/2, \\ y_j + \lambda/2, & y_j < -\lambda/2, \\ 0, & |y_j| \le \lambda/2. \end{cases}$$

Proof. The objective separates: ridge minimises $(y_j - \beta_j)^2 + \lambda\beta_j^2$; lasso minimises $(y_j - \beta_j)^2 + \lambda|\beta_j|$, whose subgradient condition $-2(y_j - \beta_j) + \lambda s_j = 0$, $s_j \in \partial|\beta_j|$, gives the soft-thresholding rule. ■

Ridge shrinks every coefficient by the same proportion; the lasso subtracts a constant and truncates at zero (soft thresholding).

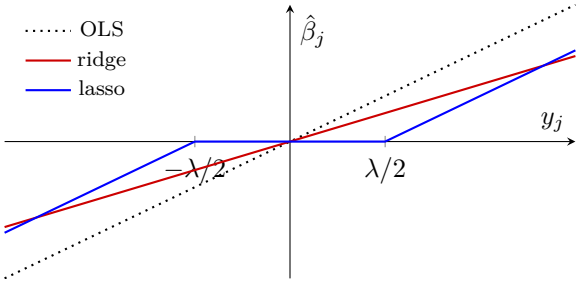


Figure 2: Orthonormal design: ridge rescales y_j by $1/(1+\lambda)$ (here $\lambda = 0.6$), the lasso soft-thresholds it, setting $|y_j| \le \lambda/2$ exactly to zero.

3.5 Dense versus sparse truth

When all 45 predictors matter (dense), ridge slightly beats the lasso; when only 2 of 45 matter (sparse), the lasso wins clearly. Neither universally dominates. Expect the lasso to do better when the response depends on relatively few predictors; but that number is never known a priori, so CV decides on a given data set.

3.6 Selecting the tuning parameter

Choose a grid of λ values, compute the CV error for each, pick the minimiser, and refit on all observations with the chosen λ.

4 Theoretical View in High Dimensions

Consider $Y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{n\times p}$, possibly $p \gg n$. OLS is unstable or not identifiable; regularisation makes estimation possible by imposing structure: **ridge** assumes β dense but small in ℓ_2; the **lasso** assumes β sparse or approximately sparse. Their relative performance depends on the geometry of the true β.

Theorem: lasso rate under sparsity (informal)

Let β be s-sparse, $\|\beta\|_0 \le s$. Under a restricted-eigenvalue (RE) condition on X and with $\lambda \asymp \sigma\sqrt{\log p/n}$ (in the normalisation $\frac{1}{2n}\mathrm{RSS} + \lambda\|\beta\|_1$),

$$\|\hat\beta - \beta\|_2^2 \lesssim \frac{\sigma^2 s \log p}{n}$$

with high probability. Hence consistent estimation is possible when $n \gg s\log p$.

The price for not knowing the support is only a $\log p$ factor compared with the oracle rate $\sigma^2 s/n$, which explains why the lasso is effective for sparse signals.

Why design assumptions are needed. Sparsity alone is not sufficient: if columns of X are highly correlated, different sparse vectors produce similar fitted values. Guarantees require conditions such as restricted eigenvalue, compatibility, or (stronger) restricted isometry; exact support recovery needs even stronger assumptions — an irrepresentable condition and a beta-min condition ($\min_{j\in S}|\beta_j|$ above the noise level).

Ridge, $\hat\beta^{\text{ridge}} = \arg\min\{\|Y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\}$, does not exploit sparsity but stabilises estimation under multicollinearity; it is favourable when β is dense with small ℓ_2 norm, where shrinking all coefficients slightly beats enforcing sparsity.

Effective complexity. For sparse vectors $B_0(s) = \{\beta : \|\beta\|_0 \le s\}$ the complexity is essentially $s\log p$; for dense vectors $B_2(R) = \{\beta : \|\beta\|_2 \le R\}$ it is governed by ℓ_2 geometry. The lasso is adapted to ℓ_1-type sparsity, ridge to ℓ_2-type regularity.

5 Dimension Reduction Methods

5.1 General idea

Construct $M < p$ linear combinations

$$Z_m = \sum_{j=1}^{p} \phi_{mj} X_j, \qquad m = 1, \dots, M,$$

and fit $Y = \theta_0 + \sum_{m=1}^{M} \theta_m Z_m + \varepsilon$ by least squares. Each Z_m is a direction in feature space; the choice of ϕ_{mj} determines the method. Since

$$\sum_{m=1}^{M} \theta_m Z_m = \sum_{j=1}^{p} \beta_j X_j, \qquad \beta_j = \sum_{m=1}^{M} \theta_m \phi_{mj},$$

dimension reduction constrains β to an M-dimensional subspace, which reduces variance (at the cost of bias if β lies outside it).

5.2 Principal components regression

PCA defines the directions: the first principal component is the normalised linear combination with the largest variance; the second has the largest variance subject to being uncorrelated with the first, and so on (eigenvectors of the sample covariance matrix). With many correlated variables we replace them by a few components capturing their joint variation.

Example: advertising data, population and ad spending in 100 cities

$Z_1 = 0.839(\texttt{pop} - \overline{\texttt{pop}}) + 0.544(\texttt{ad} - \overline{\texttt{ad}})$, with $0.839^2 + 0.544^2 = 1$. The first PC is the line that minimises the sum of squared perpendicular distances to the points; its scores z_{i1} are strongly related to both pop and ad, while the second-PC scores are only weakly related.

Assumption: the directions in which $X_1, \ldots, X_p$ vary most are the directions associated with Y. When the response depends on the first few PCs, PCR does well; when it depends on many predictors in a sparse way, PCR needs many components. PCR with $M = p$ is least squares; PCR is close in spirit to ridge (ridge shrinks low-variance directions smoothly, PCR drops them). On the Credit data, the CV error is lowest near $M = 10$, i.e. little reduction.

Standardisation: generally standardise before PCA; otherwise high-variance variables dominate the components and the measurement scale affects the final PCR model. If all variables share the same unit one may skip it.

Limitation: the directions are found *unsupervised* — the response does not supervise them — so the directions that best explain X need not be best for predicting Y.

5.3 Partial least squares

PLS also builds $Z_1, \ldots, Z_M$ as linear combinations and fits least squares, but chooses them *supervised*: directions that approximate the predictors *and* relate to the response.

Theorem: the first PLS direction

With standardised X and centred Y,

$$w_1 = \arg\max_{\|w\|=1} \operatorname{Cov}(Xw, Y)^2 \;\propto\; X^\top Y,$$

i.e. ϕ_{1j} is proportional to the simple-regression coefficient of Y on X_j; $Z_1 = Xw_1$.

Proof. $\operatorname{Cov}(Xw, Y) = \frac{1}{n} w^\top X^\top Y$, and by Cauchy–Schwarz $(w^\top X^\top Y)^2 \le \|w\|^2 \|X^\top Y\|^2$ with equality iff $w \propto X^\top Y$. For standardised X_j the simple-regression slope of Y on X_j is $x_{(j)}^\top Y / n$. ■

Algorithm: PLS (sequential construction)

Standardise X and Y; set $X^{(0)} = X$, $Y^{(0)} = Y$. For $m = 1, \ldots, M$:

1. direction $w_m \propto X^{(m-1)\top} Y^{(m-1)}$;
2. component $Z_m = X^{(m-1)} w_m$;
3. regress and update residuals $Y^{(m)} = Y^{(m-1)} - \hat{\theta}_m Z_m$;

4. deflate the predictors $X^{(m)} = X^{(m-1)} - Z_m p_m^\top$, $p_m = X^{(m-1)\top} Z_m / (Z_m^\top Z_m)$ (each column is replaced by its residual after regressing on Z_m).

So $Y = \theta_1 Z_1 + E_1$, then $E_1 = \theta_2 Z_2 + E_2$, and so on. The components are mutually uncorrelated. On the advertising data, the first PLS direction puts more weight on `ad` than the first PC, since `ad` is more related to sales. In practice PLS often performs no better than ridge or PCR: supervision reduces bias but can increase variance.

5.4 Theoretical comparison

Method	Assumption on β	Mechanism
lasso	sparsity	ℓ_1 constraint
ridge	small ℓ_2 norm	shrinkage
PLS	low-dimensional structure	iterative projection

PLS does not assume sparsity but that β lies in a low-dimensional subspace. With $\Sigma = \mathrm{Cov}(X)$ and $c = \mathrm{Cov}(X, Y)$, the m-component PLS estimator lies in the Krylov subspace

$$\mathcal{K}_m(\Sigma, c) = \mathrm{span}\{c, \Sigma c, \ldots, \Sigma^{m-1} c\},$$

(it equals least squares restricted to that subspace, i.e. m steps of conjugate gradients on the normal equations), and it is consistent if $\beta \in \mathcal{K}_m(\Sigma, c)$, i.e. when the signal has low-dimensional structure. Since $\beta = \Sigma^{-1} c$, this holds e.g. when c lies in the span of m eigenvectors of Σ.

Summary

- Best subset is accurate but expensive; forward/backward stepwise search $1+p(p+1)/2$ models; the final size is chosen by C_p, AIC, BIC, adjusted R^2 or CV.
- Ridge shrinks all coefficients (always beats OLS for some λ); the lasso shrinks and selects; standardise first and choose λ by CV.
- In high dimensions the lasso attains $s \log p / n$ under design conditions; ridge suits dense signals.
- PCR regresses on unsupervised principal components; PLS uses supervised directions maximising covariance with Y (Krylov subspace).